Laws of Logarithms

Exam code: 9MA0

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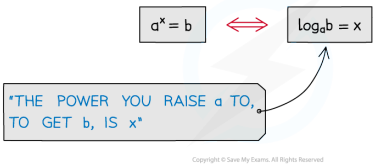
Laws of logarithms

What are the laws of logarithms?



- There are many laws or rules of indices, for example
  - $a^m \times a^n = a^{m+n}$
  - $(a^m)^n = a^{mn}$
- There are equivalent laws of logarithms (for a > 0)
  - $\log_a xy = \log_a x + \log_a y$
  - $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$
  - $\log_a x^k = k \log_a x$

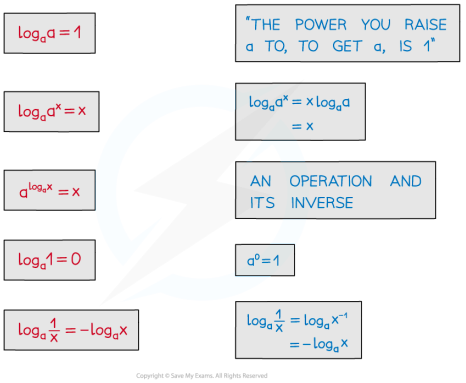
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- There are also some particular results these lead to
  - $\log_a a = 1$
  - $\log_a a^x = x$
  - $a^{\log_a x} = x$
  - $\log_a 1 = 0$
  - $\log_a\left(\frac{1}{x}\right) = -\log_a x$

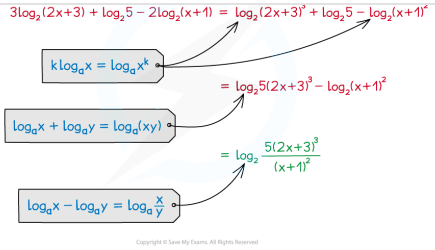


- Two of these were seen in the notes Logarithmic Functions
- Beware ...
  - ... **$\log(x + y) \neq \log x + \log y$**

- Results apply to **ln** too
  - $\ln x \equiv \log_e x$
  - In particular $e^{\ln x} = x$ and $\ln(e^x) = x$

How do I use the laws of logarithms?

- Laws of logarithms can be used to ...
  - ... simplify expressions
  - ... solve logarithmic equations
  - ... solve exponential equations

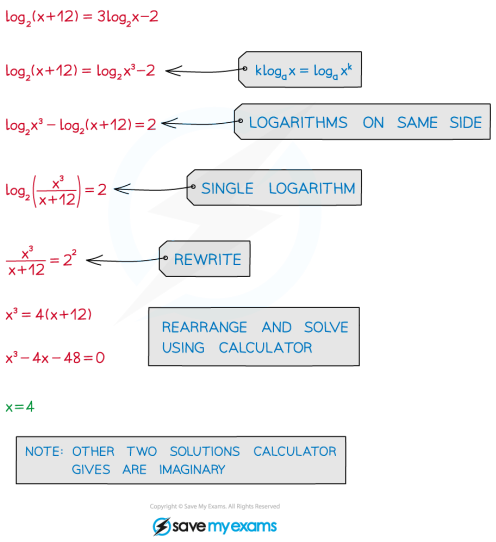


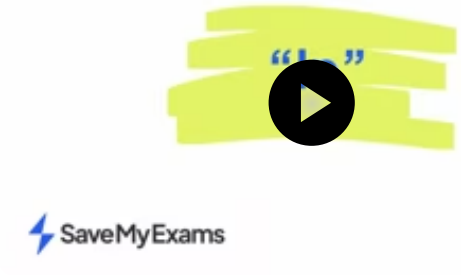
Examiner Tips and Tricks

- Remember to check whether your solutions are valid
  - log (x+k) is only defined if x > -k
  - You will lose marks if you forget to reject invalid solutions

Worked Example

Solve the equation $\log_2(x + 12) = 3\log_2 x - 2$





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"ln"

What is ln x?

- **ln** is a function that stands for **natural logarithm**
- It is a logarithm where the **base** is the constant "**e**"
  - $\ln x \equiv \log_e x$
  - It is important to remember that **ln** is a function and **not a number**

What are properties of ln x?

- Using the **definition** of a **logarithm** you can see
  - $\ln 1 = 0$
  - $\ln \mathrm{e} = 1$
  - $\ln \mathrm{e}^x = x$
  - $\ln x$ is only defined for positive *x*

- As ln is a logarithm you can use the **laws of logarithms**
  - $\ln a + \ln b = \ln(ab)$
  - $\ln a - \ln b = \ln\left(\frac{a}{b}\right)$
  - $n \ln a = \ln(a^n)$

How are e^x and ln x inverses of each other?

- The functions e^x and $\ln x$ are **inverses** of each other
  - If $\mathrm{e}^{\mathrm{f}(x)} = \mathrm{g}(x)$ then $\mathrm{f}(x) = \ln \mathrm{g}(x)$
  - If $\ln \mathrm{f}(x) = \mathrm{g}(x)$ then $\mathrm{f}(x) = \mathrm{e}^{\mathrm{g}(x)}$

- If your equation involves "e" then try to get all the "e" terms on one side
  - If "e" terms are multiplied, you can add the powers
    - $\mathrm{e}^x \times \mathrm{e}^y = \mathrm{e}^{x+y}$
    - You can then apply ln to both sides of the equation

  - If "e" terms are added, try transforming the equation with a substitution
    - For example: If $y = \mathrm{e}^x$ then $\mathrm{e}^{4x} = y^4$
    - You can then solve the resulting equation (usually a quadratic)
    - Once you solve for *y* then solve for *x* using the substitution formula

single term
- If you have $\ln f(x) = \ln g(x)$ then solve $f(x) = g(x)$
- If you have $\ln f(x) = k$ then solve $f(x) = e^k$

Worked Example

Examiner Tips and Tricks

- Always simplify your answer if you can
  - for example, $\frac{1}{2}\ln 25 = \ln\sqrt{25} = \ln 5$
  - you wouldn't leave your final answer as $\sqrt{25}$ so don't leave your final answer as $\frac{1}{2}\ln 25$

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